Sen2Coral : Algorithm Theoretical Basis Document and Data Processing Model

Module A9: Physical Modelling

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1.1	22/06/2018	CG, VB		3.1	Clarification of performance assumptions

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Figure 1. (a) Input image showing field stations that might be used for validating the products (b) output map of bottom depth with mask applied to land, (c) output map of three-substrate type map. 3

Figure 2. – Table showing the list of the variables mentioned in the text (in column one the number of the equation where the variable appears). The spectral variables, which depend on wavelength, have the λ-dependence included in the symbol; otherwise, they are scalars. The units for the most common variables are indicated between square-brackets. For each variable it is noted whether the variable is fixed (F) or may be optimized (O). 5

1. INTRODUCTION

1.1. Product Overview

1.1.1. Product Description

The ability to accurately estimate water column depth over large areas and/or in remote locations is directly relevant to environmental management, exploration, defense and research applications.

The product described in this document is developed according to the state-of-the-art of spectra inversion of semi-analytical (SA) bio-optical modelling. This method relies on some form of spectrum matching of modeled reflectance spectrum using a set of bathymetry, bottom classification, and water column IOPs values, optimized to match the image Rrs(λ) spectra. In particular the method:

- it builds on Lee et al. (1998, 1999) equations;
- It has included the quality control procedure based on additional outputs of the integrated physics based mapping approach (e.g. Substrate Detectability Index) and the characteristics of the instrument (environmental noise equivalent reflectance/radiance) as in SAMBUCA (Brando et al., 2009);
- It has assumed the a relative linear mixed distribution of up three different substrates as in BOMBER (Giardino et al., 2012);
- It is developed with a user-friendly interface similar to BOMBER (Giardino et al., 2012);
- It makes use of a relax constraints hypothesis (e.g. Petit et al. 2017) for modelling the contribution of the three substrates in bottom reflectance.

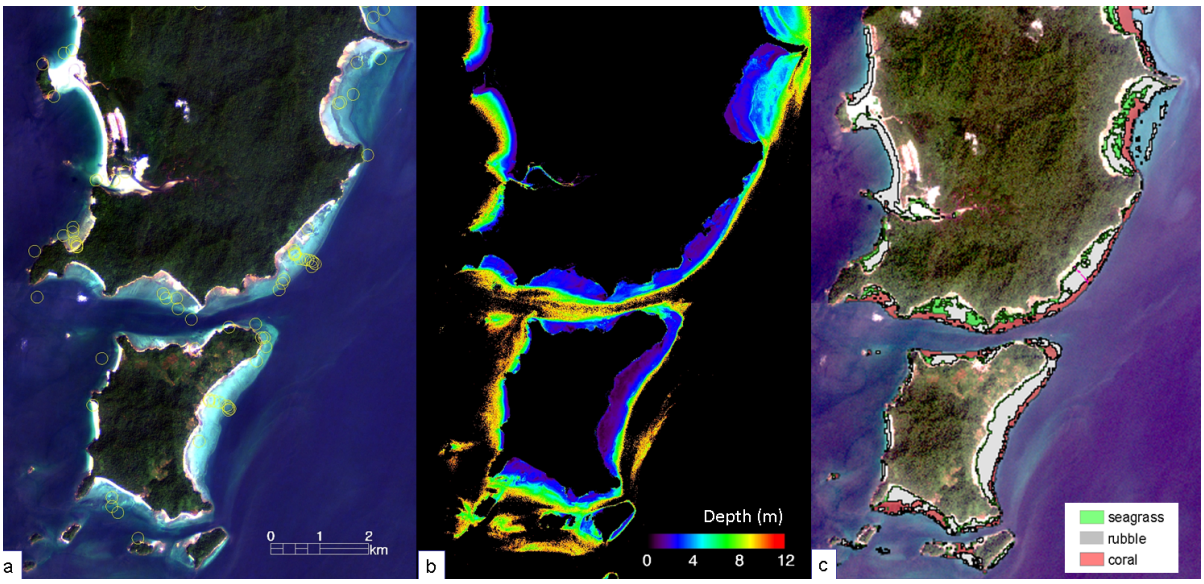

Figure 1. (a) Input image showing field stations that might be used for validating the products (b) output map of bottom depth with mask applied to land, (c) output map of three-substrate type map.

1.1.2. Product Requirements

The requirement for the product is a source image in units of remote sensing reflectance (Rrs(λ), above water, in sr-1) with a minimum of five bands in water penetrating wavelengths, i.e. in visible wavelengths, not the near infra-red (NIR).

The Rrs (the λ-dependency might be omitted for sake of clarity) image bands must be accurately treated to correct for the radiometric effects. In particular, the atmospheric effects including secondary source of noise such as adjacency effects.

Knowledge of the optical water type (e.g. clear versus turbid waters) and of the coastal environment (coral bays versus reefs) is welcome to initialize the model. In-situ bathymetry data and GPS locations with a description of substrates is desirable for validation data set. However, these information points are not part of the algorithm and so are not further discussed in this document.

1.2. Satellite Instrument Description

The product is not specific to any instrument and is applicable to any source of passive optical remotely sensed images that contains one or more bands in the visible wavelengths. This includes, Sentinel 2 MSI, Landsat 8 OLI, and all known high spatial resolution satellite sensors.

2. ALGORITHM DESCRIPTION

2.1. Processing Outline

The method to be implemented will follow that of Lee et al. (1999), Brando et al. (2009), Giardino et al. (2012) and Petit et a. (2017) which relies on spectral inversion of bio-optical modelling.

Bio-optical models are mathematical equations, which relate radiometric variables (e.g., radiance reflectance, remote sensing reflectance, irradiance reflectance) observed above or below the water surface to the inherent optical properties (IOPs) of the optical active constituents or to the concentrations of water quality parameters. Phytoplankton, suspended matter and coloured dissolved organic matter (also known as yellow substances or gelbstoff) are the main elements influencing the optical water properties and the radiative transfer in the water column. In case of shallow waters, the bottom contribution to the water leaving radiance is also relevant and bio-optical models must include this component as well.

The purpose of this ATBD is to present SWAMpy (Shallow Water semi-Analytical Model in python), an add-on tool for SNAP. SWAMpy implements the inversion of a bio-optical model and make use of optimization techniques to assess bottom properties from Sentinel-2 images. In particular, SWAMpy generates maps of bottom depth and bottom distribution of three substrate classes. SWAMpy relies on the python implementation of SAMBUCA (i.e. Brando et al., 2009) made available by CSIRO, and implements also other recent features implemented by other authors (e.g. Giardino et al., 2012) and Petit et al.,2017) .

In SWAMpy, the above water remote sensing reflectance Rrs(λ), is expressed as a function of the subsurface radiance reflectance r0-(λ), which is approximated as the sum of contributions from the water column r0-(λ)C and the bottom r0-(λ)B:

$$R_{\mathrm{rs}}(\lambda) = \frac{\Pi r_0^-(\lambda)}{(1-\gamma r_0^-(\lambda))}, \quad (1)$$

$$r_0^-(\lambda) = r_0^-(\lambda)^{\mathrm{C}} + r_0^-(\lambda)^{\mathrm{B}} = r_0^-(\lambda)^{\mathrm{dp}}\left(1 - A_0 e^{\left(-\left(\frac{1}{\cos(\vartheta_{\mathrm{w}})}+D_{\mathrm{u}}^{\mathrm{C}}\right)\kappa H\right)}\right) + A_1\rho(\lambda)e^{\left(-\left(\frac{1}{\cos(\vartheta_{\mathrm{w}})}+D_{\mathrm{u}}^{\mathrm{B}}\right)\kappa H\right)}, \quad (2)$$

$$r_0^-(\lambda)^{\mathrm{dp}} = \left(g_0 + g_1 u(\lambda)^{g_2}\right)u(\lambda), \quad (3)$$

with $$u(\lambda) = \frac{b_{\mathrm{b}}(\lambda)}{a(\lambda)+b_{\mathrm{b}}(\lambda)}, \quad (4)$$

where Π and γ in equation 1 are model parameters which take into account the effects of air-water interface on the transmission of radiance. r0-(λ)dp in equations 2 and 3 is the contribution of optically deep waters in sr-1. The variables cos(θw) and H in equation 2 are the cosine of the underwater sun zenith angle and the bottom depth, respectively. The quantity u(λ) in equation 4 is a function of the two IOPs: total absorption coefficient a(λ) and the total backscattering coefficient bb(λ), which are

modelled as functions of concentrations of water constituents (cf. equations 8-14). For the definition of the model parameters g0, g1, g2, A0, A1, Dcu, DBu and k and for a more detailed description of the bio-optical model the user can refer to Lee et al. (1998, 1999).

The bottom albedo ρ(λ) in equation 2 is expressed as:

$$\rho(\lambda) = b_0\rho_0(\lambda) + b_1\rho_1(\lambda) + b_2\rho_2(\lambda) \quad (5)$$

With b0+b1+b2=1 or 0.5 <b0+b1+b2 <2, (6)

where bi=0,1,2 represent the relative distribution of three different substrate albedo ρ(λ))i=0,1,2 defined by the user.

The bio-optical model design implemented in SWAMpy allows the user to describe IOPs according to two models found in literature and suitable for coral environments: Lee Stocking Island and Moreton Bay (Dekker et al., 2011). Users can prepare their own IOPs definition by editing the input XML files.

The optimization process is then performed through the minimization of a target function δ which measures the spectral distance between modelled and image-derived Rrs(λ), using different kind of error function. There are seven variables for equation 7 (details in section 3.1): chlorophyll concentration (CHL), suspended particulate matter concentration (SPM), colored dissolved organic matter (aCDOM(440)), bottom depth (H), bottom distribution b0, b1 and b3 (if b2 is a function of b0 and b1 according to equation 6, the variables are reduced to six) of pre-defined albedo classes.

Starting from initial values of the unknowns, the predictor-corrector procedure changes these values computing an error function error_f, at each step. The algorithm repeats this process until error_f reaches a minimum. Values corresponding to the error_f minimum represent the best (optimum) set for the unknown variables. The algorithm can use four different error functions (LSQ, f, alpha and alpha_f). A complete list of all variables is given in Table 1.

$$\begin{cases} LSQ = \sqrt{\sum_{i=i}^{N}(r_{rs}^{\mathrm{mod}\,el}(\lambda_i) - r_{rs}^{obs}(\lambda_i))^2} \\ f = \sqrt{\sum_{i=i}^{N}\frac{1}{nedr(\lambda_i)}(r_{rs}^{\mathrm{mod}\,el}(\lambda_i) - r_{rs}^{obs}(\lambda_i))^2} \\ alpha = ar\cos\left(\frac{\sum_{i=i}^{N}\frac{1}{nedr(\lambda_i)} * (r_{rs}^{\mathrm{mod}\,el}(\lambda_i) * r_{rs}^{obs}(\lambda_i))}{\sqrt{[\sum_{i=1}^{N}(\frac{1}{nedr(\lambda_i)} r_{rs}^{\mathrm{mod}\,el}(\lambda_i))^2]} * \sqrt{[\sum_{i=1}^{N}(\frac{1}{nedr(\lambda_i)} r_{rs}^{obs}(\lambda_i))^2]}}\right) \\ alpha_f = f * \alpha \end{cases} \quad (7)$$

Figure 2. – Table showing the list of the variables mentioned in the text (in column one the number of the equation where the variable appears). The spectral variables, which depend on wavelength, have the λ-dependence included in the symbol; otherwise, they are scalars. The units for the most common variables are indicated between square-brackets. For each variable it is noted whether the variable is fixed (F) or may be optimized (O).

Equation	Symbol	Definition	F/O
1	$R_{rs}(\lambda)$	Remote sensing reflectance [sr^{-1}]	O
1	Π, γ	Model parameters to account the effects of air-water interface on the transmission of radiance	F
1, 2, 7	$r_0^-(\lambda)$	Subsurface radiance reflectance [sr^{-1}]	O
2	$r_0^-(\lambda)^C$	Subsurface radiance reflectance the water column [sr^{-1}]	O
2	$r_0^-(\lambda)^B$	Subsurface radiance reflectance the bottom [sr^{-1}]	O
2, 3	g_0, g_1, g_2, A_0, A_1, D^C_u, D^B_u, k	Parameters required by the "Deep Waters" and "Shallow Waters" models according to Lee et al. (1998, 1999).	F

2	ϑ_w	Underwater sun zenith angle [deg]	F
2	H	Bottom depth [m]	O
2, 3	$r_0^-(\lambda)^{dp}$	Subsurface radiance reflectance for optically deep waters [sr^{-1}]	O
2, 5	$\rho(\lambda)$, $\rho(\lambda)_{i=0,1,2}$	Bottom albedo of the *i-th* substrate (i=0,1,2) [-]	F
3, 4	$u(\lambda)$	Variable used to relate $r_0^-(\lambda)^{dp}$ to $a(\lambda)$ and $b_b(\lambda)$ [-]	O
4, 8	$a(\lambda)$	Total absorption coefficient [m^{-1}]	O
4, 12	$b_b(\lambda)$	Total backscattering coefficient [m^{-1}]	O
5, 6	$b_{i=0,1,2}$	Relative distribution of the *i-th* substrate ($b_0+b_1+b_2=1$)	O
7	δ	Target function of the optimisation process [-]	O
7	$\hat{R}_{rs}(\lambda)$	Image-derived remote sensing reflectance [sr^{-1}]	F
7	*LSQ, f, alpha and alpha_f*	Error associated to the optimisation [-]	O
8	$a_w(\lambda)$	Absorption coefficient of pure water [m^{-1}]	F
8, 9	$a_\Phi(\lambda)$	Absorption coefficient of phytoplankton [m^{-1}]	O
8, 10	$a_{NAP}(\lambda)$	Absorption coefficient of non-algal particles [m^{-1}]	O
8, 11	$a_{CDOM}(\lambda)$	Absorption coefficient of CDOM [m^{-1}]	O
9, 14	CHL	Chlorophyll concentration [mgm^{-3}]	O
9	$a^*_\Phi(\lambda)$	Specific absorption coefficient of phytoplankton [m^2mg^{-1}]	F
10, 17	NAP	Non-algla-particulate concentration [gm^{-3}]	O
10	S_{NAP}	Slope of the exponential curve describing $a_{NAP}(\lambda)$ [-]	O
11	$a_{CDOM}(440)$	Absorption coefficient of CDOM at 440 nm (i.e. CDOM concentration) [m^{-1}]	O
11	S_{CDOM}	Slope of the exponential curve describing $a_{CDOM}(\lambda)$	F
13	$b_{bw}(\lambda)$	Backscattering coefficient of pure water [m^{-1}]	F
14	$b_{b\Phi}^*(\lambda)$	Chlorophyll-specific phytoplankton backscattering [m^2mg^{-1}]	F
12, 14	$b_{b\Phi}(\lambda)$	Backscattering coefficient of phytoplankton [m^{-1}]	O
12, 16	$b_{bNAP}(\lambda)$	Backscattering coefficient of NAPs [m^{-1}]	O
16	$b_{bNAP}^*(\lambda)$	Specific backscattering coefficient of NAPs [m^2g^{-1}]	F

2.2. Processing Steps

SWAMpy is entirely programmed in Python 3.4 (optimized for WinPython and Anaconda) and runs as add-on tool of the SNAP. The optimization algorithm is based on the Sequential Least Squares Programming (SLSQ), a procedure available in the Python Scipy package, useful for the minimization of a scalar function of one or more variables. The graphic user interface (GUI) is realized through java widget tools.

Once installed, a new module appears in the SNAP--> Optical--> Sen2Coral Processing menu bar and selecting SWAMpy button, a properly sheet window with several option menus and buttons appears on the screen. The menu manage all feature necessary to: 1) parameterize the bio-optical model, 2) select input image and output products, and 3) define values involved in the calculations and optimization processes.

Input Image - This menu allows the user to choose the input image file to perform the optimization on (Fig. 1a). The input image must be atmospherically corrected and expressed into Rrs values in sr-1 units.In the processing parameters window, the user can specify if the Rrs is relative to above or in-water by checking the specific box; in the first case it will be converted into in-water rrs by the algorithm.

Output Products - Here the user can choose where to save the file containing the products retrieved by the optimization process (Fig. 1a). The output product provides for each pixel of the input image: depth, SDI, Kd, the fractions of the three substrates, the reflectance of the bottom (r_sub) and the error function (error_f), used in the optimization process. The user can choose the latter from the processing parameters window. Four error functions are available: LSQ, f, alpha and alpha_f (defined in the algorithm description).

SIOP xml file: The specific inherent optical properties (SIOPs) must be chosen by the user selecting a proper xml file from the processing parameters window (SIOP_xml_file). This .xml files can be created using the xmltodict and SAMBUCA libraries, both available in Python3.4.

Absorption Coefficients - The total absorption coefficient (cf. equation 4) is defined by the formula:

$a(\lambda)=a_w(\lambda)+a_\Phi(\lambda)+a_{NAP}(\lambda)+a_{CDOM}(\lambda)$ (8)

where $a_w(\lambda)$, $a_\Phi(\lambda)$, $a_{NAP}(\lambda)$ and $a_{CDOM}(\lambda)$ are the absorption coefficients of pure water, phytoplankton, non-algal particles and coloured dissolved organic matter, respectively.

The phytoplankton absorption coefficient $a_\Phi(\lambda)$ is modelled according to Brando and Dekker 2003 and Keller et al., 2001 :

$$a_\phi(\lambda) = a_\phi^*(\lambda)CHL \quad (9)$$

The non-algal particle absorption coefficient aNAP(λ) is defined as suggested by Babin et al. (2003):

$$a_{NAP}(\lambda) = a_{NAP}^*(\lambda)NAP = a_{NAP}^*(\lambda_{0NAP})e^{-S_{NAP}(\lambda-\lambda_{0NAP})}NAP \quad (10)$$

The absorption coefficient of coloured dissolved organic matter $a_{CDOM}(\lambda)$ is modelled as:

$$a_{CDOM}(\lambda) = a_{CDOM}^*(\lambda)CDOM = a_{CDOM}^*(\lambda_{0CDOM})e^{-S_{CDOM}(\lambda-\lambda_{0CDOM})} \quad (11)$$

S_{cdom}, S_{nap}, $a_{nap}(\lambda_{0NAP})$, $a_{cdom}(\lambda_{0CDOM})$ are provided by the xml file. So, in order to modify the absorption by NAP and CDOM, the user can simply modifies the values of such quantities in the xml file (opening it from suitable programs, like XML Notepad).

a_w and a_ϕ* are explicitly provided by the xml file. As for the constants, involved in the a_{cdom} and a_{nap} calculations, the user can modify the values of the spectra and the corresponding wavelenghts directly in the file, but this can be a really hard work. So, when the user need to modify these spectra, it's reccomended to create a new xml file, using the xmltodict and SAMBUCA libraries, available for Python 3.4.

Backscattering Coefficients - The total backscattering coefficient is provided by the sum of three components:

$bb(\lambda)=bb_w(\lambda)+bb_\Phi(\lambda)+bb_{NAP}(\lambda)$ (12)

where $bb_w(\lambda)$, $bb_\Phi(\lambda)$ and $bb_{NAP}(\lambda)$ are the backscattering coefficients of pure water, phytoplankton and non-algal particles, respectively.

The backscattering coefficient of pure water is modelled using the formula:

$$bb_w(\lambda) = \frac{0.00194}{2}\left(\frac{\lambda_{0bw}}{\lambda}\right)^{4.32} \quad (13)$$

The backscattering coefficient of phytoplankton $bb_\Phi(\lambda)$ is defined as in Brando and Dekker (2003):

$$bb_\phi(\lambda) = bb_\phi^*(\lambda)CHL \quad (14)$$

where the chlorophyll-specific phytoplankton backscattering bb_Φ*(λ) is calculated by the following equation:

$$bb_\phi^*(\lambda) = bb_\phi^*(\lambda_{0b\phi})\left(\frac{\lambda_{0b\phi}}{\lambda}\right)^{Y_\phi} \quad (15)$$

The backscattering coefficient of non-algal particle $bb_{NAP}(\lambda)$ is modelled as follow:

$$bb_{NAP}(\lambda) = bb_{NAP}^*(\lambda)NAP \quad (16)$$

Where the specific backscattering coefficient for the Non-Algal Particles is:

$$bb^*_{NAP}(\lambda) = bb^*_{NAP}(\lambda_{0bNAP})(\frac{\lambda_{0bNAP}}{\lambda})^{Y_{NAP}} \quad (17)$$

The slopes, the reference wavelengths and the corresponding specific backscattering values are stored in the xml file and they can be directly modified by the user.

Bottom Albedo xml file - Also the bottom albedo of the three substrates is contained in the SIOP xml file. To modify these spectra it is recommended to create a new xml file with Python.

Bounds and Starting Values xml file -The bounds and the starting values are stored in the xml_parameters_file. Here the bounds can be modified by the user, changing the values of the items contained in p_min and p_max (in order: chl, cdom, nap, depth and the substrates fractions). For all the parameters, except for depth, the starting values are simply the mid points between their minimum and their maximum values. The starting points for the depth is defined by the formula:

$$h_0 = 10^{\log_{10}(\frac{h_{\max}}{h_{\min}})} \quad (18)$$

SLSQP Arguments - In the processing parameters window we can also modify the optimization process arguments. We already mentioned the possibility of choosing different error functions, used in the SLSQP process. Besides that, we can also decide to use the relaxed constrains hypothesis, by checking the specific box.

Sensor filters, signal noise and rrs bands: the sensor filters and the signal noise (NEDR) are contained in the sensor_filter_xml file. Both of them can be modified directly in the file or using the SAMBUCA and dictotxml libraries. The wavelenght range that will be used in the computation can be modified in the processing parameters window.

The user must choose wisely the sensor filters and wavelengths range because:

1) the sensor filters contained in the file must refer to the sensor of the input image.

2) the algorithm, in order to select the bands used in the computation, acquires the values of the spectral wavelength of each of them and selects only the ones included in the chosen range. So first of all, the user must check if all the spectral wavelength values of the bands of the interest are present, right clicking on the bands and selecting "properties". If the values are absent or mistaken, the user must modify them. On the contrary the spectral wavelength fields of other bands, such radiances of irradiances, must be empty or outside the range of interest, because the algorithm doesn't discern the rrs from other quantities.

3) the number of central spectral wavelengths of the sensor filters must be the same of the number of image bands included in the range selected by the user.

2.3. Algorithm Input

- An image bands in Rrs units
- SIOP
- Subtrates reflectance
- Sensor filters
- NEDR (environmental noise equivalent reflectance)
- Wavelenght range
- Bounds
- Type of rrs
- Choice of the Error function

2.4. Theoretical Description

2.4.1. Physical Description

The potential to estimate water column depth, constituents of the water column, and benthic cover types from remote sensing data have been discussed for over two decades. Significant attention has been paid to empirical approaches (Clark et al. 1987; Lyzenga 1981; Philpot 1989) that although might be location- or data-limited (Lee et al. 2001; Stumpf 1987) they still provide useful product over large areas (cf. atdb a8_empirical_bathymetry). More recently, semi-analytical (SA) and radiative transfer-based forward models and SA or look-up table based spectrum matching (LUT) inverse-models have been developed having several advantages (e.g. repeatability, transferability) for mapping shallow water environments, presuming that a remote sensing image has already been corrected for atmospheric and air/water interface effects (Dekker et al. 2007).

In case of forward-modelling the approach is typically based on matching of image spectra to a precomputed library of spectra corresponding to different combinations of depth, bottom type, and water properties. This approach was developed by Louchard et al. (2003) and Mobley (2005) and applied to coral reef mapping in Lesser and Mobley (2007).

A different approach is used in case of inverse-modelling. Lee et al. (1999) first developed a semi-analytical model for shallow water remote sensing based on radiative transfer. Lee et al. (2001) used an inversion optimization approach to simultaneously derive water depth and water column properties from hyper-spectral data in coastal waters. Adler-Golden et al. (2005) present an algorithm similar to that of Lee et al. (2001). However, it makes the simplifying assumption of constant water optical properties within the scene. Mcintyre et al. (2006) presented an application of the (Lee et al. 2001) inversion modeling approach to clear waters that included a quantitative comparison of model-derived depth with high resolution multi-beam acoustic bathymetry data.

Several authors recently extended the method developed by Lee et al. (1999) and Lee et al. (2001) by incorporating linear unmixing of the benthic cover. Brando et al. (2009) developed of a quality control procedure based on additional outputs of the integrated physics based mapping approach and the characteristics of the instrument within SAMBUCA. Giardino et al. (2007) integrated a SA inversion model with a fractional cover considering up to three substratum types in the BOMBER code developed with user-friendly GUIs useful for educational purposes. Gege (2004) developed WASI and WASI-2D codes foe also including an analytic model of downwelling irradiance that allows wavelength dependent modelling of sun glint and sky glint at the water surface. Dekker et al. (2011) compared the performances of five methods referred to as HOPE, BRUCE, SAMBUCA, CRISTAL, and ALLUT. HOPE, BRUCE, and SAMBUCA perform spectrum matching via a nonlinear search that determines the optimal parameter values for their SA model, while CRISTAL performs spectrum matching by searching through a precomputed database of Rrs spectra. ALLUT is a hybrid application that merges these two approaches: it uses an adaptive LUT method for creating an effective LUT for parameterizing the SA model and then applies spectrum matching. Then, since most of these methods do not consider the distribution of targeted parameters within the inversion process, thereby implicitly assuming that any parameter value in the estimation range has the same probability, Petit et al. (2017), proposed to regularize the objective functions of estimation methods by introducing local prior knowledge on the parameters of interest.

2.4.2. Mathematical Description

The mathematical description has been covered in the previous section.

2.5. Algorithm Output

The output is an image with bands of bottom depth (H) and bottom distribution (b0, b1 and b2) of three substrate classes. An additional map is provided with a measure of the error (err) associated to

the inversion. These images have all bands with the same spatial size (samples and lines) and geometry as of the input image.

2.6. Performance Estimates

2.6.1. Test Data Description

- Description: SWAMpy algorithm applied to a preprocessed Sentinel-2 product.
- Input Parameters:

Parameter	Value
source	mini_lizard_2016_03_23_tdg_cmn_u0_tau0p18_LEboa.dim
xmlpath_sensor	swampy_s2_5_bands_filter_nedr.xml
xmlpath_parameters	swampy_s2_parameters_depth30.xml
xmlpath_siop	swampy_s2_6_siop_Lizard.xml
error_name	alpha_f
opt_method	SLSQP
min_wlen	400.0
max_wlen	750.0
above_rrs_flag	true
shallow_flag	true
relaxed_cons	true

- Expected output: mini_LIZARD_SAMBUCA_ALL.dim
- Acceptance criteria: The 95% of the pixels in each band differs by less than the 2% from the expected output.

2.6.2. Sensor Effects

The algorithm is primarily affected by atmospheric and adjacency effects; the environmental noise equivalent reflectance/radiance is part of the optimization process and might reduce residual noise due to sensor, atmosphere and adjacency.

2.6.3. Retrieval Errors

It is provided by the error function.

2.7. Practical Considerations

2.7.1. Numerical Computation Considerations

The algorithm is complex and the computational process might require time. However, the use of application of the algorithm to a multi-spectral data such as Sentinel-2 might not be so critical.

2.7.2. Programming and Procedural Considerations

None.

2.7.3. Quality Assessment and Diagnostics

The test data set and output are compared to the outputs of the implementation. The values should be identical within the tolerance of numerical errors and minor implementation details.

2.8. Verification

The test data set and output are compared to the outputs of the implementation. The values should be identical within the tolerance of numerical errors and minor implementation details.

3. ASSUMPTIONS AND LIMITATIONS

3.1. Performance Assumptions

The performances of SWAMpy strictly depend on the quality of input files. Reliable estimates are expected when both Rrs/rrs values of image data and model parameters (both spectral files and scalars) are appropriate (e.g. accurate glint removal and atmospheric correction), as shown for other shallow water semi-analytical models (e.g. Goodman et al., 2008; Brando et al., 2009; Dekker et al., 2011; Botha et al., 2016). Particularly, it could happen that after the atmospheric correction, a residual reflectance is present in the near infrared bands on deep waters. In this case, the algorithms will not provide valid values, since this input spectrum is typical of shallow water, proding depth retrievals that do not represent at all the truth.

3.2. Potential Improvements

Extension to other water types and substrate types.

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